



20n-02
MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY
JAIPUR

22MET101: Introduction to Mechanical Systems

End-term Exam – April 30, 2026

B. Tech. II Semester (A, B, C, D, E and F Batches)

MM: 50

Time: 150 min

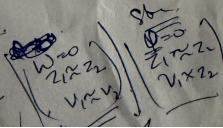
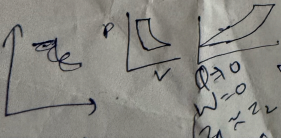
Instructions: All the questions are compulsory. Marks and course objectives are indicated against each question. Draw sketches/labelled diagrams wherever necessary. Attempt the questions in the given sequence. Use of steam tables & scientific calculator is permitted.

SECTION – A ($5 \times 4 = 20$ marks)

- Q1. With the help of a neat sketch, briefly explain the following machining operations:
(a) Counterboring, (b) Countersinking, (c) Boring, (d) Reaming [4M, CO2]
- Q2. Explain the following mechanical properties of engineering materials:
(a) Strength (b) Toughness (c) Hardness (d) Malleability [4M, CO3]
- Q3. List various types of thermodynamic systems along with a representative engineering example. State the conditions under which a pressure cooker behaves as a closed system and an open system, respectively. [4M, CO3]
- Q4. Distinguish between a heat pump and a refrigerator? A cold storage facility is to be maintained at -5°C while the outdoor air is at 35°C . The refrigeration load is estimated to be 29 kW. If the actual COP of the refrigeration plant is one-third of the ideal value for the plant operating between same temperatures, find the power required to produce the desired cooling. [4M, CO3]
- Q5. Steam enters the nozzle, operating at steady state at 30 bar and 350°C with negligible velocity and exits at 15 bar and a velocity of 500 m/s. If the mass flow rate is 2.22 kg/s, determine the exit temperature of steam. [4M, CO3]

SECTION – B ($5 \times 6 = 30$ marks)

- Q6. State at least six differences between a four-stroke and a two-stroke engine. [6M, CO3]
- Q7. Draw a block diagram showing the key components of a Simple Vapour Compression Refrigeration (VCR) System. Also, state the thermodynamic process corresponding to each component of the system and show the cycle on T-s diagram. List any two engineered devices which operate on VCR cycle. [6M, CO3]
- Q8. Draw a labelled diagram of a Center Lathe indicating various components and describe its working principle. [6M, CO3]
- Q9. An engine operates on air-standard Diesel cycle. The pressure and temperature at the beginning of compression are 100 kPa and 27°C . The compression ratio is 18 and the heat added per kg of air is 1850 kJ. Determine (a) maximum pressure during the cycle, (b) maximum temperature during the cycle and (c) thermal efficiency of the cycle. Assume $\gamma = 1.4$, $c_p = 1.005$ kJ/kg. K. Also, show the cycle on p-v and T-s diagrams. [6M, CO3]
- Q10. A cylindrical pressure vessel, with diameter and height equal to 80 cm and 160 cm, contains steam at 12 bar with a dryness of 0.95. Using a pressure release valve, some steam is blown off until the pressure drops to 6 bar. Steam pressure drops further to 4 bar due to spontaneous cooling of the vessel. Treating steam blow-off as a throttling process, determine (a) Quantity of blown-off steam (b) Steam quality after spontaneous cooling of vessel (c) Heat lost by the steam due to cooling. [6M, CO3]



Boiler 20 bar
Steam turbine
B



Department of Mechanical Engineering

End-Term Examination

Course title: Introduction to Mechanical Systems

Course code: 22MET101

Programme/ Semester: B. Tech./ I

Session: Jan – May, 2025
(Even Semester)

Max. Marks: 50

Time Duration: 150 minutes

Date: 01/05/2025

- Note: 1. All the questions are compulsory. Marks are indicated against each question.
2. Use of scientific calculator and steam tables is permitted. Programmable/ graphical calculator is not allowed.
3. Draw neat labeled diagrams, sketches and graphs wherever required. All answers should be to the point.
4. Answer scripts with matching solution, partly or wholly, will be summarily rejected without prior information.
5. Attempt the questions in the given sequence. Answers of the numerical problems should be accompanied with S.I. units.

Section A (5 x 4 = 20 marks)

- (a) Find the specific volume & specific enthalpy of a steam with 50% dryness at a pressure of 50 kPa. 4
(b) A closed rigid vessel contains wet steam at 0.5 MPa. If the wet steam comprises 125 grams and 7.5 grams of liquid and vapour, respectively, determine the volume of the vessel.
- Manoj, his older brother Ankit, and their father sleep in separate, air-conditioned rooms. On a given night of summers, Manoj sets his air-conditioner to 23°C, Ankit to 20°C, and their father to 18°C. Arrange the three family members in the decreasing order of energy-efficiency of their room air-conditioning and support your claim by computing a suitable performance parameter. Also, draw a block diagram showing the four essential components of an air-conditioner and briefly explain the operating principle. Assume missing data, if any. 4
- Show the Air-standard diesel cycle on the two property diagrams, pressure-volume and temperature entropy, respectively. If the compression ratio equals 15 and the heat addition occurs over 6% of the piston's stroke, obtain the thermal efficiency of the engine. 4
- Distinguish between a refrigerator and a heat pump. If both operate between same heat reservoirs, which one performs more efficiently? Support your claim with a mathematical proof. 4
- List atleast four machining operations, along with a simple sketch, which cannot be performed on a general purpose Center Lathe. Furthermore, suggest a machine tool suitable for executing each listed operation. 4

Section B (5 x 6 = 30 marks)

- Explain the operating principle of oxy-acetylene gas welding. List atleast four welded joints and show each with a neat diagram. 6
 - In a textile industry, steam with a high flow velocity is required for treating the raw cotton fibers. For that purpose, steam at 10 bar and 350 °C is made to expand in a nozzle until its pressure reduces to 1.2 bar. Determine the exit velocity and temperature, if the flow velocity at nozzle inlet is 50 m/s. 6
 - Draw a labelled schematic of an evaporative air-cooler and briefly discuss its working. Enumerate atleast two differences between an air-cooler and a residential air conditioner. 6
 - An ammonia refrigerator produces 10 metric tonnes of ice at 0°C in a day. While the feed water is supplied at 30 °C, the temperature range of vapour compression cycle is 25°C to -15°C. The ammonia is dry saturated at the end of compression and it is saturated liquid at the exit of condenser. If the actual COP is 60% of the theoretical value, find the power requirement of the compressor. Saturation properties of the refrigerant are tabulated below. Take latent heat of fusion for the water equal to 335 kJ/kg K. Note: Subscript 'l' refers to saturated liquid and the subscript 'v' refers to saturated vapour. Assume missing data, if any. 6
- | T (°C) | h _l (kJ/kg) | h _v (kJ/kg) | s _l (kJ/kg K) | s _v (kJ/kg K) |
|--------|------------------------|------------------------|--------------------------|--------------------------|
| 25 | -644.86 | 520.57 | 6.105 | 10.016 |
| -15 | -831.45 | 481.52 | 5.438 | 10.526 |
- A certain quantity of air at a pressure of 1 bar and temperature 70°C is compressed adiabatically until the pressure rises to 7 bar in an engine that runs on Otto cycle. If 465 kJ of heat per kg of air is added during heat addition process, show the cycle on p-V and T-s diagrams and determine (a) Compression ratio of the engine (b) Temperature at the end of compression (c) Temperature at the end of heat addition. For air, $c_p = 1.0$ kJ/kg K, $c_v = 0.706$ kJ/kg K. 6



Department of Mechanical Engineering

End-Term Examination

Course title: Introduction to Mechanical Systems

Course code: 22MET101

Programme/ Semester: B. Tech./ I

Session: Aug – Dec, 2024
(Odd Semester)

Max. Marks: 50

Time Duration: 150 minutes

Date: 04/12/2024

- Note: 1. All the questions are compulsory. Marks are indicated against each question.
2. Use of scientific calculator and steam tables is permitted. Programmable/ graphical calculator is not allowed.
3. Draw neat labeled diagrams, sketches and graphs wherever required. All answers should be to the point.
4. Answer scripts with matching solution, partly or wholly, will be summarily rejected without prior information.
5. Attempt the questions in the given sequence. Answers of the numerical problems should be accompanied with S.I. units.

Section A (5 x 3 = 15 marks)

1. Apart from cooling the room air, list atleast two additional functions of an air-conditioner aimed at improving human comfort. 3
2. Derive the numerical equivalent of 1 TR in S.I. units. If an air conditioner is rated at 1.5 TR, what is its cooling capacity in kW? 3
3. Determine whether the working fluid is a sub-cooled liquid, saturated liquid, wet vapour, dry saturated, or superheated in the following cases (a) Steam at 175 °C and 5 bar (b) Steam at 8 bar and 22 °C 3
4. Which thermodynamic cycle governs the operation of Spark Ignition Engines? Show the cycle on both pressure-volume and temperature-entropy diagrams. 3
5. List atleast three machining operations performed on a drilling machine and show each with a simple diagram. 3

Section B (7 x 5 = 35 marks)

6. Compression ratio and maximum temperature of an air standard diesel cycle is 18 and 2800 K respectively. If pressure and temperature at the start of compression stroke is 1 bar and 300 K respectively, determine the cut-off ratio and thermal efficiency of the cycle. Take $\gamma=1.4$ 5
7. Why is it necessary to have a suitable cooling system for an uninterrupted operation of internal combustion engines? Draw a labeled schematic of a water cooled system and briefly explain its functioning. 5
8. A rigid vessel of 0.0076 m³ capacity contains 0.05 kg steam at 15 bar. Predict the steam temperature. If the vessel undergoes cooling, determine the temperature when steam becomes dry saturated. If cooling continues until pressure drops to 11 bar, determine the steam quality. 5
9. Explain the operating principle of shielded metal arc welding. List atleast three merits as well as demerits of arc welding compared to oxy-acetylene gas welding. 5
10. An ice-cream factory daily produces 5 metric tonnes of ice-cream at 0 °C. If the COP of the plant is 3.0, mechanical efficiency of the compressor motor is 90 % and the initial temperature of liquid hot cream is 38 °C, determine (i) Rate of heat removal from the hot cream (ii) Power requirement of the compressor motor. Take specific heat of liquid hot cream equal to 4186 J/kg-°C, and latent heat of freezing is 334.5 kJ/kg. 5
11. List atleast five lathe operations. Briefly explain their application and show each with a simple diagram. 5
12. Briefly explain the operating principle, list key components, and draw a neat diagram of a Domestic Refrigerator. 5

Shalini

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Mechanical Engineering

First Year B.Tech (A,B,C,D,E,F)
End Term Exam

Even Semester: 2023-24

Max Marks: 50

Date :06-05-2024

SUBJECT: Introduction to Mechanical Systems(22MET-101)

- Note: 1. All the questions are compulsory. Marks are indicated against each questions.
2. Use of Scientific calculator is permitted. Programmable/graphical calculator is not allowed.
3. Draw neat labelled diagrams, sketches and graphs whenever required. All answers should be to the point.
4. Answer script with matching solution, partly or wholly, will be summarily rejected without prior information.
5. Attempt the questions in the given sequence. Answer of the numerical problems should be accompanied with S.I units.

✓ Que 1. Suggest a suitable gear drive for the following cases of driver and driven shafts (i) Shafts are intersecting (ii) shafts are non-coplanar and non-intersecting (iii) shafts at an angle (iv) shafts are parallel (v) Gear of infinite radius (5)

✓ Que 2. A diesel engine has a compression ratio of 15 and heat addition at constant pressure takes place at 6% of stroke. Find the air standard efficiency of the engine. Draw the P-V and T-S diagram of diesel cycle also. (5)

✓ Que 3. Explain the principle of arc welding and its applications. List the equipments and tools used with their functions in arc welding with diagram. (5)

✓ Que 4. A carnot refrigerator requires 1.25 KW per ton of refrigeration to maintain a region at low temperature of -40°C . Determine (i) COP Carnot refrigerator (ii) Higher temperature of the cycle and (iii) Heat delivered and COP when this device is used as heat pump. Draw the block diagram of heat pump, heat engine and refrigerator. (5)

✓ Que 5. Two pulleys, one 450 mm dia. and other 200 mm dia are on parallel shafts 2 m apart are connected by a crossed belt. Find the length of belt required. What power can be transmitted by a belt when the larger pulley rotates at 200 rpm and maximum permissible tension in the belt is 1KN? Given the coefficient of friction between the belt and pulley is 0.25. (5)

or

✓ In a flat belt drive prove that $T_1/T_2 = e^{\mu\theta}$, Where T_1 is tension in tight side, T_2 is tension in slack side, μ is coefficient of friction, θ is angle of lap in radian. (5)

✓ Que 6. Explain the function of five hand tools used in foundry with neat sketch? (5)

✓ Que 7. What is Air Conditioning. What do you mean by ton of refrigeration. Briefly explain the operating principle, list key components, and draw a neat diagram of window air conditioner. (5)

✓ Que 8. A storage tank closed at both ends contains steam at a pressure of 8 bar and temperature of 200°C . It has an internal volume of 0.652 m^3 . If the pressure in the storage tank falls to 3 bar due to heat loss from the steam pipe, find the mass of steam in the pipe and then quality and temperature of the steam at the final state. (5)

✓ Que 9. Describe the operation of the two stroke in I C Engine in which only air is taken during the suction stroke. How it is different from petrol engine. (5)

✓ Que 10. Draw a neat diagram of centre lathe indicating various key component. List any five lathe operations. (5)



MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
(Institution of National Importance under NITs Act, Established by Govt. of India)

मालवीय राष्ट्रीय प्रौद्योगिकी संस्थान जयपुर
JLN Marg, Jaipur-302017 (India)

Department of Mechanical Engineering

End-Term Examination

Course title/ code : Basics of Mechanical Engg. (MET 101)

Programme/ semester: B. Tech./ First

Session: Jul-Dec, 2019 (Odd semester)

Max. Marks: 50

Duration: 150 minutes

Date: 3/12/19

- Note: 1. All the questions are compulsory. Marks are indicated against each question.
2. Use of scientific calculator is permitted. Programmable/ graphical calculator is not allowed.
3. Answers of the numerical problems should be accompanied with S.I. units.
4. Try to attempt the questions in the given sequence. All answers should be to the point.
5. Draw neat labeled diagrams, sketches and graphs wherever required

Section A (12 marks)

1. Derive the numerical equivalent of 1 TR in S.I. units. Briefly explain the difference between refrigeration and air conditioning. 2
2. What is the difference between drilling, boring & reaming? Show with the help of neat diagram. 2
3. Give atleast two advantages and two disadvantages of belt drives over gear drives. 2
4. Center distance between two flat belt pulleys is 5 m and diameters of the pulleys are 1.6 m and 2.2 m respectively. If a crossed belt is desired to be used, find the belt length. 2
5. List atleast four properties of the moulding sand. Briefly explain the role of any two of them. 2
6. What is the operating principle of domestic refrigerator? Show the thermodynamic cycle on temperature-entropy diagram. Name any one refrigerant used for charging domestic refrigerators nowadays. 2

Section B (28 marks)

7. Draw the schematic diagram, list main components and briefly explain the working of a commercial ice plant. Also, show the thermodynamic cycle on temperature-entropy diagram. 4
8. With a neat diagram, explain the construction and working of a two stroke engine. 4
9. A rigid vessel of 0.0076 m^3 capacity contains 0.05 kg steam at 15 bar. Predict the steam temperature. If the vessel undergoes cooling, determine the temperature when steam becomes dry saturated. If cooling continues until pressure drops to 11 bar, determine the steam quality. 4
10. Explain the operating principle of shielded metal arc welding. List atleast three merits as well as demerits of arc welding compared to oxy-acetylene gas welding. 4
11. List atleast four lathe operations. Briefly explain their application and show each with a simple and neat diagram. 4

12. A 5 TR R-12 refrigeration system has suction temperature of -5°C . The condensation takes place at 32°C and there is no sub-cooling of the refrigerant in the refrigeration cycle. The vapour at the evaporator exit is dry saturated. Assuming isentropic compression, determine (i) Mass flow rate requirement of the refrigerant (ii) Power requirement of the compressor (iii) COP of the refrigeration system. Saturation properties of R-12 are given below. Take specific heat of superheated vapour, $c_p = 0.615 \text{ J/kg K}$. 4

Temp ($^{\circ}\text{C}$)	Pressure (bar)	h_l (kJ/kg)	h_v (kJ/kg)	s_l (kJ/kg K)	s_v (kJ/kg K)
32	7.85	135	366.1	1.106	1.542
-5	2.61	195.3	350.6	0.983	1.562

13. A Carnot refrigerator consumes power equal to 2.25 kW/ TR to maintain a cabin at a low temperature -20°C . Determine (i) COP of the refrigerator (ii) Temperature of heat rejection. If the same device is used for heating applications, determine (i) Rate of heat supplied to the cabin (ii) COP 4

Section C (10 marks)

14. a) Suggest suitable gears for the following cases of driver and driven shafts (i) Shafts are mutually parallel and rotating in the same direction (ii) Shafts are non-parallel as well as non-intersecting 2
- b) An open belt drive is used to transmit 20 kW of power. The linear velocity of the belt is 2 m/s and the angle of lap on the smaller pulley is 160° . If the coefficient of friction is 0.25, determine the change in power transmitted in the following cases. (i) Initial tension in the belt decreases by 10% (ii) Lap angle increases by 10% while speed and tension on the tight side remains unchanged. 8

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
DEPARTMENT OF MECHANICAL ENGINEERING

End-Term Exam (B.Tech., 2nd Sem.; 2014-15)

Max-Marks: 40

Date: 04/05/2015

Time: 2 hrs (8-10 AM)

MET-101 Basic Mechanical Engineering

Q.1	(a) Differentiate the role of mountings and accessories of a boiler. State few examples of each. (b) Differentiate water tube boilers from fire tube boilers (5 points only)	(5)
Q.2	Sketch labelled diagram of assembled mould using two piece split pattern. What is the role of vent holes, riser and gate-runner	(5)
Q.3	Explain following terms: (a) pattern (b) casting (c) adhesiveness (d) green sand (e) porosity	(5)
Q.4	What is the significance of powers transmitting drive systems and list various types available. Also, write significant features of each one.	(5)
Q.5	A V-belt of 6.4 cm^2 cross-section has a groove angle of 30° and the angle of lap of 165° . The weight of the V-belt is 1.4 kg/meter run. The maximum safe stress is 3 N/mm^2 . Calculate the power transmitted at the velocity of 20 m/sec . Take $\mu = 0.1$.	(5)
Q.6	With neat sketch explain (any two): (a) spinning over lathe (b) form milling (c) End milling	(5)
Q.7	A Diesel engine has a compression ratio of 15 and heat addition at constant pressure take place at 6% of stroke. Find the air standard efficiency of the engine.	(5)
Q.8	Sketch and explain (a) Shielded Metal Arc Welding (b) Ice-plant	(5)

End-Term Exam 1st Semester

Malaviya National Institute of Technology Jaipur

Department of Mechanical Engineering

2013-14

Max-Marks: 40

Name: _____

Batch and Group:

Time: 2 hrs [8 to 10 AM]

ME-110 Basic Mechanical Engineering

Question-1 Sketch neat and properly labelled diagrams of:

 $(3 \times 6 = 18)$

- Section-1 Sketch neat and properly labelled diagrams of: (3×6=18)
- a) Explain with neat sketch the various machining operations that can be performed on lathe machine tools.
- b) Using split wooden pattern, explain in detail the casting procedure.
- c) Write brief note on V-belt and design.

Question-2

(5+5=10)

- a) 2 kW of power is transmitted by an open belt drive. The linear velocity of the belt is 2 m/s. The angle of lap on the smaller pulley is 160° . The coefficient of friction is 0.25. Determine the effect on power transmission in the following cases: (i) Initial tension in the belt is increased by 8% (ii) Initial tension in the belt is decreased by 8% (iii) Angle of lap is increased by 8% by the use of an idler pulley, for the same speed and the tension on the tight side (iv) coefficient of friction is increased by 8% by suitable dressing to the friction surface of the belt. (5+5=10)
- b) A four cylinder petrol engine of 250 mm bore and 375 mm stroke works on Otto cycle. The clearance volume is 0.01052 m^3 . The initial pressure and temperature are 1 bar and 47°C . If the maximum pressure is limited to 25 bar, find the following: (i) The air standard efficiency of the cycle (ii) The mean effective pressure for the cycle.

Question-3

 $(3 \times 4 = 12)$

- a) Sketch neat and labelled diagram of (i) Stages of working of four stroke diesel & petrol engine, and (ii) Gear terminology. (3×4=12)
- b) Explain the required properties of moulding sand.
- c) Explain the principle of arc and gas welding.

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MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
MECHANICAL ENGINEERING DEPARTMENT
FIRST YEAR B.TECH. (ODD SEMESTER 2012-13)
END TERM EXAM (29/11/2012)
SUB: BASIC MECHANICAL ENGINEERING

exam-support.in

MM: 40

Time: Two Hrs

Note: Attempt all the questions.

Q.1	(A) A shaft rotating at 200 r.p.m. drives another shaft at 300 r.p.m and transmits 6 kW through a belt. The belt is 100mm wide and 10mm thick. The distance between the shafts is 4m. The smaller pulley is 0.5m in diameter and the coefficient of friction between the belt and pulley is 0.3. Calculate the stress in the belt, if it is a cross belt drive.	6
	(B) Derive an expression for the ratio of tensions in flat belt drive.	4
	(C) How gears are classified?	2
Q.2	(A) What machining operations can be done on lathe? Shows these operations by neat diagrams.	4
	(B) Describe with the help of a suitable diagram the principle parts of horizontal milling machine.	3
Q.3	Find the enthalpy and entropy of steam when the pressure is 2 MPa and the specific volume is 0.09 m ³ /kg.	3
Q.4	(A) Define triple point and critical point of water.	2
	(B) Explain oxyacetylene gas welding with a suitable sketch.	4
Q.5	Write short notes on: (any Four) (A) Brazing (B) Cochran Boiler (C) Spark Plug (D) Fuel Pump (E) Summer Air Conditioning System	3*4 =12

Malaviya National Institute of Technology Jaipur
Department of Mechanical Engineering

End Term Exam (B.Tech., 2nd Semester: 2013-14)

Max-Marks: 40

Date: 05/05/2014

ME-110 Basic Mechanical Engineering

Time: 2 hrs (8-10 AM)

Attempt all questions neatly, each carries equal marks of five

- A V-belt of 6.4 cm² cross-section has a groove angle of 30° and the angle of lap of 165°. The weight of the V-belt is 1.4 kg/meter run. The maximum safe stress is 3 N/mm². Calculate the power transmitted at the velocity of 20 m/sec. Take $\mu = 0.1$.
- Define the term foundry and explain important properties required in moulding sand.
- State assumptions and derive expression for the ratio of tensions in case of V-belt drive system.

OR

What are the objectives of power transmission and enlist the means of achieving them. How slip differs from creep phenomenon.

- Compare Otto cycle from Diesel cycle.

OR

Explain the working of 2-stroke diesel engine. Compare air cooling system from water cooling system.

- Define machine tools with examples. How different feed mechanisms are obtained in case of lathe machine tool.
- Differentiate mounting from accessories in case of boilers. Explain critical point and triple point of water with their significance.
- Explain the concept of centrifugal tension in belts and how it affects power transmission. Explain the concept of soldering and its procedure. Also explain the principle of arc welding.



47 11/12